

PROFILE

AI/ML Engineer with a strong internship at Bacancy Technology where I delivered end-to-end machine-learning solutions, built production-grade retrieval-augmented generation pipelines, and automated lead-generation workflows. Proficient in Python, modern LLM stacks (LangChain, LangGraph, Gemini), computer-vision, MLOps tooling (Docker, FastAPI, GitHub Actions, AWS SageMaker) and data-driven product development. Demonstrated ability to quantify impact—cut inference latency by 78%, improve lead conversion by 12% and reduce data-processing time by 60%—while maintaining rigorous code quality and monitoring.

EXPERIENCE

- AI/ML Engineer Intern — Bacancy Technology

Jan 2026 - Apr 2026
- Designed and shipped a Dockerized image-classification microservice (PyTorch CNN) to AWS SageMaker, achieving a 78% reduction in inference latency (350 ms → 75 ms) and supporting 4x higher request throughput.
 - Implemented a CI/CD pipeline with GitHub Actions that automatically builds Docker images, runs unit/integration tests (90% coverage), and performs blue-green deployments, cutting release cycle time from 2 weeks to 2 days.
 - Built a monitoring stack (Prometheus + Grafana) to track model drift, latency, and error rates, enabling proactive alerts that reduced out-of-spec predictions by 30%.
 - Developed end-to-end computer-vision pipelines using OpenCV and MediaPipe for defect detection, incorporating data augmentation and automated preprocessing that lifted validation accuracy from 82% to 89%.
 - Collaborated with backend engineers to expose model inference via FastAPI, wrote OpenAPI specs, and integrated authentication, resulting in a production-ready API consumed by three internal services.
 - Authored comprehensive documentation and conducted knowledge-transfer sessions, ensuring the team could maintain and extend the ML service independently.

PROJECTS

- Real-Time Squid Game – Red Light Green Light

Python, OpenCV, MediaPipe, FastAPI, Docker, TensorRT
- Created a pose-tracking engine with MediaPipe that detects player movement at 30FPS, applying TensorRT optimization to achieve <20ms per frame on an NVIDIA RTX-3060.
 - Implemented game-logic server using FastAPI; containerized the service with Docker and deployed to AWS Elastic Container Service, supporting 50 concurrent players with sub-50ms response time.
 - Profiled and refactored the image-processing pipeline (cProfile) to reduce CPU usage by 40%, enabling smooth gameplay on commodity laptops.
 - Wrote a custom calibration routine that adapts lighting conditions, increasing detection reliability from 85% to 96% across varied environments.

- AI-Powered Lead Outreach Automation (n8n)

n8n, Apify, Google Sheets, OpenAI GPT-4, Python, Docker
- Engineered an end-to-end lead-generation workflow that scrapes 1,200+ company profiles daily, enriches data via Apify, and stores results in Google Sheets.
 - Integrated GPT-4 for dynamic 3-step email sequence generation, achieving a 12% lift in email open rates and a 9% increase in qualified-lead conversion.
 - Automated error handling, retry logic, and performance metrics collection; reduced manual lead-curation effort by 80% and cut average pipeline runtime from 45min to 12min.
 - Dockerized the n8n instance and set up a GitHub Actions workflow to deploy updates to a Kubernetes cluster, ensuring zero-downtime releases.

- RAG-Based YouTube Q&A Assistant

LangChain, LangGraph, FAISS, Gemini 1.5-Pro, FastAPI, Streamlit, Docker
- Built a retrieval-augmented generation pipeline that ingests YouTube transcript data, creates dense embeddings with Gemini, and stores them in FAISS for sub-200ms similarity search.
 - Fine-tuned Gemini 1.5-Pro on a domain-specific corpus (5k Q&A pairs) achieving 15% higher F1-score on intent classification compared to the base model.
 - Implemented a token-budget controller that reduced average LLM cost per query from \$0.12 to \$0.07 while maintaining answer relevance >85%.
 - Deployed the backend as a FastAPI service behind an NGINX reverse proxy, containerized with Docker, and exposed a Streamlit UI for real-time querying; monitored latency and usage with Grafana dashboards.

SKILLS

AI_ML

Python

TensorFlow/Keras

PyTorch

Scikit-learn

OpenCV

Computer Vision

NLP

LangChain

LangGraph

DEV

FastAPI

Docker

Kubernetes

AWS SageMaker

AWS ECS/EKS

Git / GitHub

GitHub Actions

CI/CD

RESTful API design

SQL

Pandas

TOOLS

MLflow

DVC

Prometheus

Grafana

n8n

Jupyter / Colab

Power BI

EDUCATION

B.E. Computer Science & Engineering (Data Science)	2022 - 2026
Vishwakarma Government Engineering College	CGPA 9.36